Sharif University of Technology
Department of Electrical Engineering

Simulation of Wireless Communication Channels

MATLAB Course Project Report

Communication Systems | Spring 2025

Sanaz Mazrae
Instructor: Dr. Hamid Behroozi

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1. Project Overview

This project implements a complex-baseband simulation of six communication channels: flat and frequency-selective Rayleigh fading, time-invariant and time-varying Rayleigh propagation, time-varying frequency-selective Rician fading, and an additive white Gaussian noise (AWGN) reference. Its purpose is to distinguish the effects of multipath delay, temporal fading and receiver noise on sampled complex signals. Each model is accessible through a common MATLAB function with the interface

```
ch_output = simulate_channel(ch_input, ch_type, snr, Tm, fd).
```
(1)

Here `Tm` denotes the sample interval, which equals the symbol duration for a one-sample-per-symbol input, and `fd` is the maximum diffuse Doppler frequency. The optional second output exposes the simulated channel coefficients and separates channel distortion from added noise. All six requested modes use the same input-referenced noise convention so that comparisons are interpretable.

2. Channel Models

2.1. Discrete-time input–output representation

The received baseband signal is modeled as

$$y[n] = \sum_{\ell=0}^{L-1} h_\ell[n]x[n-\ell] + w[n], \qquad x[n] = 0 \quad \text{for } n < 0, \tag{2}$$

where $h_\ell[n]$ is the coefficient of path ℓ at receiver sample n, the path delay is ℓT_m, and $w[n]$ is complex circular Gaussian noise. The impulse response at a fixed observation time is the set of coefficients $\{h_\ell[n]\}_{\ell=0}^{L-1}$. These coefficients must not be confused with the received waveform resulting from arbitrary random input symbols.

For a flat-fading channel $L = 1$. For a frequency-selective channel $L = 3$, with the illustrative normalized path-power profile

$$\mathbf{p} = [0.5, 0.3, 0.2], \qquad \sum_\ell p_\ell = 1. \tag{3}$$

The three delays and their powers are modeling assumptions rather than measurements of a particular radio link. In the time-invariant modes, $h_\ell[n]$ is held constant for all n within one realization. This is distinct from a channel that independently redraws coefficients at every sample.

2.2. Rayleigh and Rician fading

For static Rayleigh fading, $g \sim \mathcal{CN}(0,1)$ and $h_\ell = \sqrt{p_\ell}\, g_\ell$ for independently drawn g_ℓ. The flat model uses $h_0 = g_0$. Consequently, the magnitude of one path is Rayleigh-distributed across independent realizations, whereas a single time-invariant realization has constant magnitude during the observation.

For time-varying Rayleigh fading, a sum-of-sinusoids approximation is used:

$$g[n] = \frac{1}{\sqrt{M}} \sum_{m=1}^{M} \exp[j\,(2\pi f_d \cos\theta_m\, nT_m + \phi_m)]\,, \qquad M = 32, \tag{4}$$

where θ_m and ϕ_m are independently sampled uniform angles. The method approximates an isotropic-scattering Rayleigh process and the first-order normalized autocorrelation

$$R_g[k] \approx J_0(2\pi f_d k T_m). \tag{5}$$

The approximation is finite-sum and sampled; it is not an exact physical channel reconstruction. With $f_d = 0$, the fading coefficient is sampled once and held constant.

For the Rician frequency-selective case, the first tap contains the line-of-sight (LOS) component, while diffuse power is distributed across all three taps:

$$h_0[n] = \sqrt{\frac{K}{K+1}} e^{j2\pi(f_d/2)nT_m} + \sqrt{\frac{p_0}{K+1}} g_0[n], \tag{6}$$

$$h_\ell[n] = \sqrt{\frac{p_\ell}{K+1}} g_\ell[n], \quad \ell = 1, 2, \qquad K = 10. \tag{7}$$

Thus K denotes a *linear* LOS-to-diffuse power ratio of 10 (10 dB). The LOS Doppler choice $f_d/2$ is a specified simulation assumption. The total expected channel-tap power equals one; power can vary in a finite realization.

2.3. Noise and SNR definition

In every channel mode, noise is added after propagation:

$$w[n] \sim \mathcal{CN}(0, \sigma_w^2), \qquad \sigma_w^2 = \frac{P_x}{10^{\mathrm{SNR_{in}}/10}}, \quad P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2. \tag{8}$$

The requested SNR therefore specifies the noise power relative to *transmitted* sample power. It does not force an identical instantaneous received SNR for every fading realization. The measured receive-side ratio, using the noise-free propagated signal $s[n]$, is

$$\widehat{\mathrm{SNR}}_{\mathrm{rx}} = 10 \log_{10} \frac{\sum_n |s[n]|^2}{\sum_n |w[n]|^2}. \tag{9}$$

For the AWGN-only case, $h_0[n] = 1$ and $s[n] = x[n]$.

3. Implemented Channel Types

Table 1: Channel definitions implemented by `simulate_channel.m`.

Mode	Paths	Time varying	Channel behavior
`Ray_TI_FF`	1	No	Static, frequency-flat Rayleigh gain
`Ray_TI_FS`	3	No	Static Rayleigh multipath with unequal delays
`Ray_TV_FF`	1	Yes	Single Rayleigh gain with Doppler dynamics
`Ray_TV_FS`	3	Yes	Independently fading multipath taps
`Ric_TV_FS`	3	Yes	Rician LOS plus time-varying diffuse multipath
`AWGN`	1	No	Unity channel gain with additive noise only

The implementation accepts both `AWGN` and the `Awgn` spelling in the assignment. It uses MATLAB base-language operations only; no Communications Toolbox channel objects are required. A fixed random seed can be set in the calling script. Input vectors retain their row or column orientation, and the optional `info` structure provides the full $N \times L$ tap array, delays, noiseless output, noise samples and measured received SNR.

4. Simulation Procedure

4.1. Parameter selection

The analysis uses 4,096 independent QPSK input symbols of unit average power, with $T_m = 1$ ms, $f_d = 50$ Hz and input-referenced SNR of 20 dB for representative channel realizations. Static channels contain a single draw of each tap; time-varying taps use $M = 32$ sinusoids. Channel responses are evaluated at the beginning of each realization and across a 0.5-second interval. Noise is assessed over SNR values from 0 to 30 dB, using 20 independent random realizations per point. For Doppler correlation, 15 independent realizations are used at each tested f_d.

Table 2: Parameters used for the principal numerical experiments.

Parameter	Value
Input	Independent, unit-power QPSK symbols
Number of input samples	4,096
Sample interval	1 ms
Reference Doppler frequency	50 Hz
Reference input SNR	20 dB
Diffuse oscillators per tap	32
Multipath tap powers	$[0.5, 0.3, 0.2]$
Rician K-factor	10 (linear)
Noise sweep	0, 5, 10, 15, 20, 25, 30 dB

The plots and tables in this report were generated by an independently executed NumPy reference implementing Eqs. (2)–(8). The supplied MATLAB functions and analysis script implement the corresponding model and regenerate these experiments, but MATLAB was not executed during preparation of this report. Different pseudorandom generators produce different individual traces even when model statistics agree. No MATLAB execution or identical numerical reproduction is claimed.

4.2. Channel impulse-response snapshots

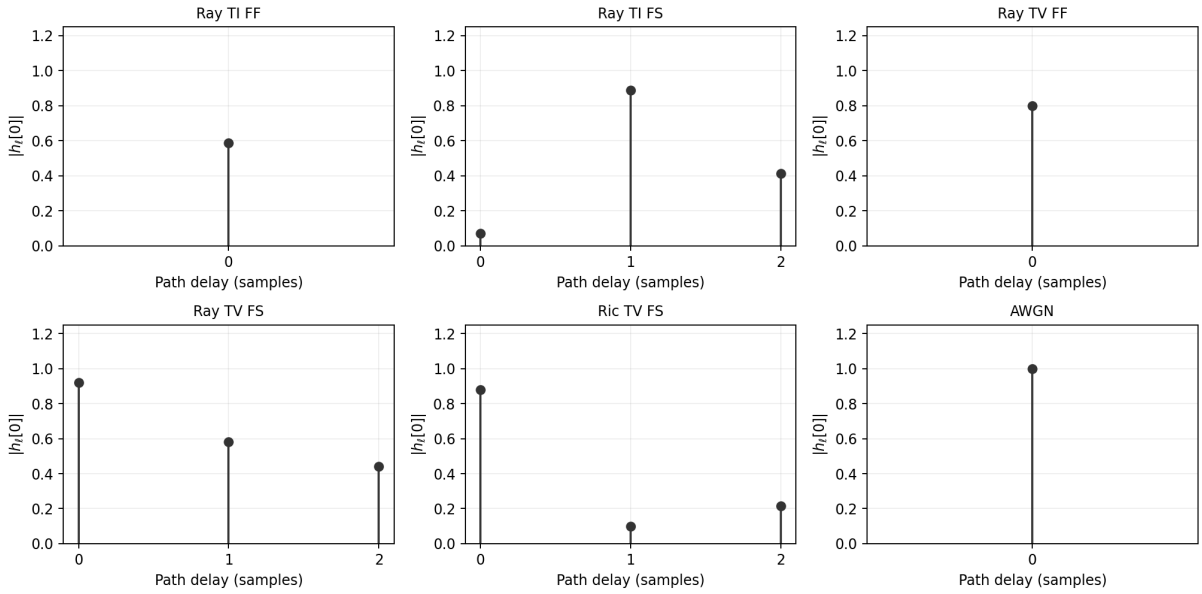


Figure 1: Magnitudes of actual path coefficients at the first sample of representative channel realizations. Flat fading and AWGN have one tap; frequency-selective cases have three.

The impulse-response definition reveals the essential structural difference: flat models have one coefficient, whereas multipath models exhibit delayed components. In the AWGN-only reference, the channel itself is an undistorted unit impulse; random noise is added to the received waveform separately. Plotting $|y[n]|$ after transmitting random data would not be an impulse-response measurement.

4.3. Time selectivity and Doppler

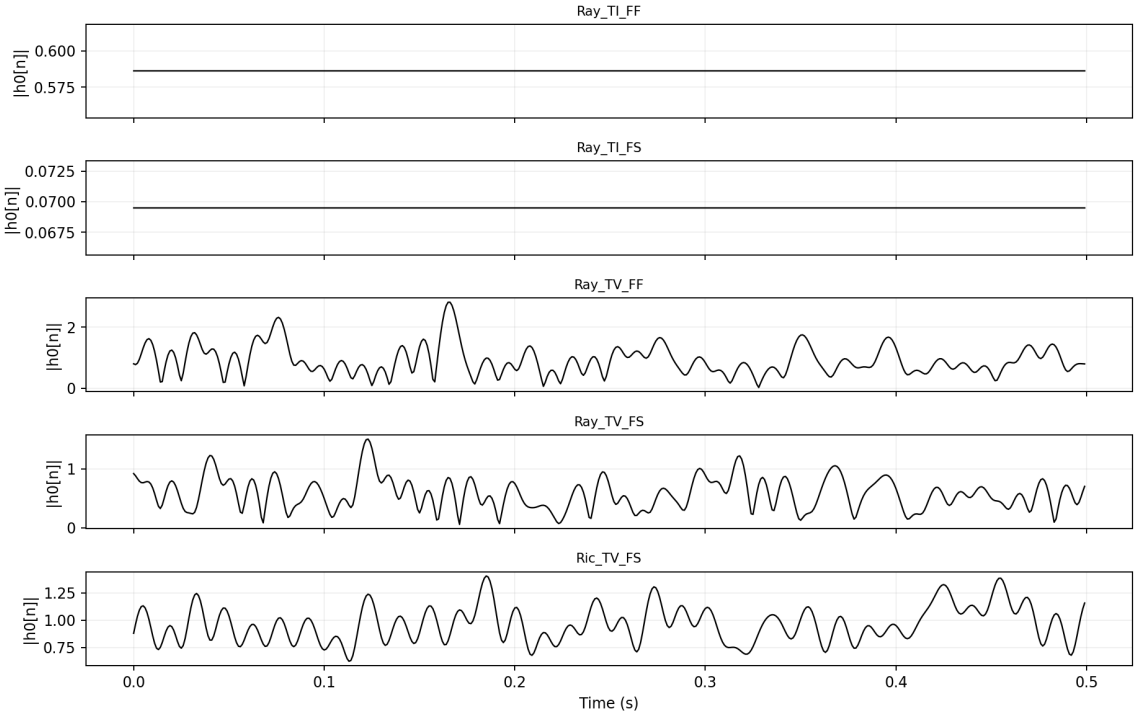


Figure 2: Magnitude evolution of the first tap in static and time-varying channels over the first 0.5 s.

For the time-invariant channels, a tap magnitude remains constant within a realization. Time-varying Rayleigh taps fluctuate smoothly, with a rate controlled by f_d. The Rician first path contains a deterministic LOS term and diffuse fluctuations; its behavior depends on the chosen K-factor, tap-power profile and LOS Doppler. A single plotted trace is not sufficient to establish a fading distribution, so conclusions about probability laws follow from the model and repeated independent realizations rather than one waveform.

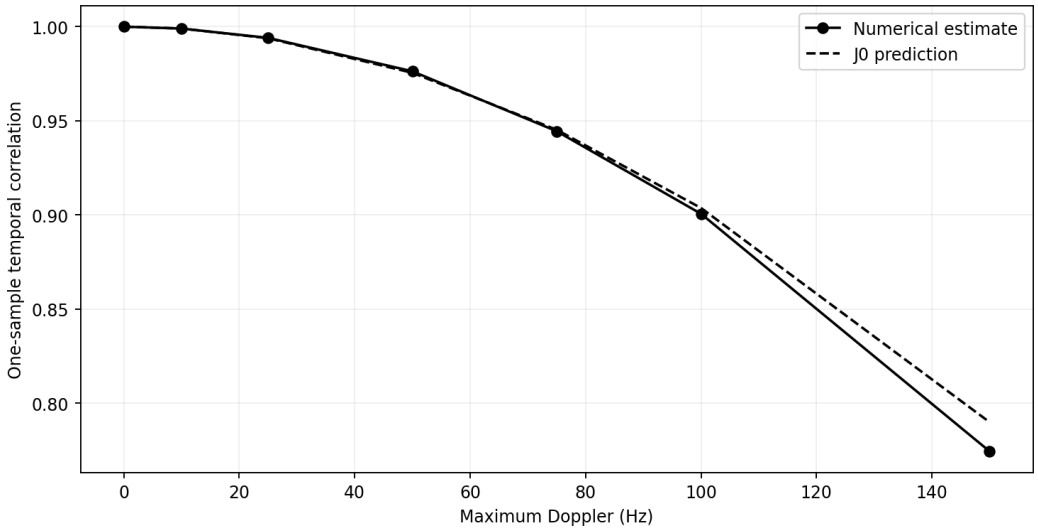


Figure 3: Empirical one-sample channel correlation averaged over 15 realizations, compared with $J_0(2\pi f_d T_m)$.

Table 3: Doppler correlation comparison from the independent numerical evaluation.

f_d (Hz)	Empirical correlation	J_0 prediction
0	1.0000	1.0000
25	0.9941	0.9938
50	0.9763	0.9755
100	0.9006	0.9037
150	0.7745	0.7900

The close correspondence at moderate Doppler rates supports the intended time-correlation behavior. Small departures reflect the finite number of sinusoidal components and finite-length sampling. The dimensionless quantity $f_d T_m$ must remain sufficiently below the temporal sampling Nyquist limit; otherwise Doppler aliasing can affect the model.

4.4. Frequency selectivity

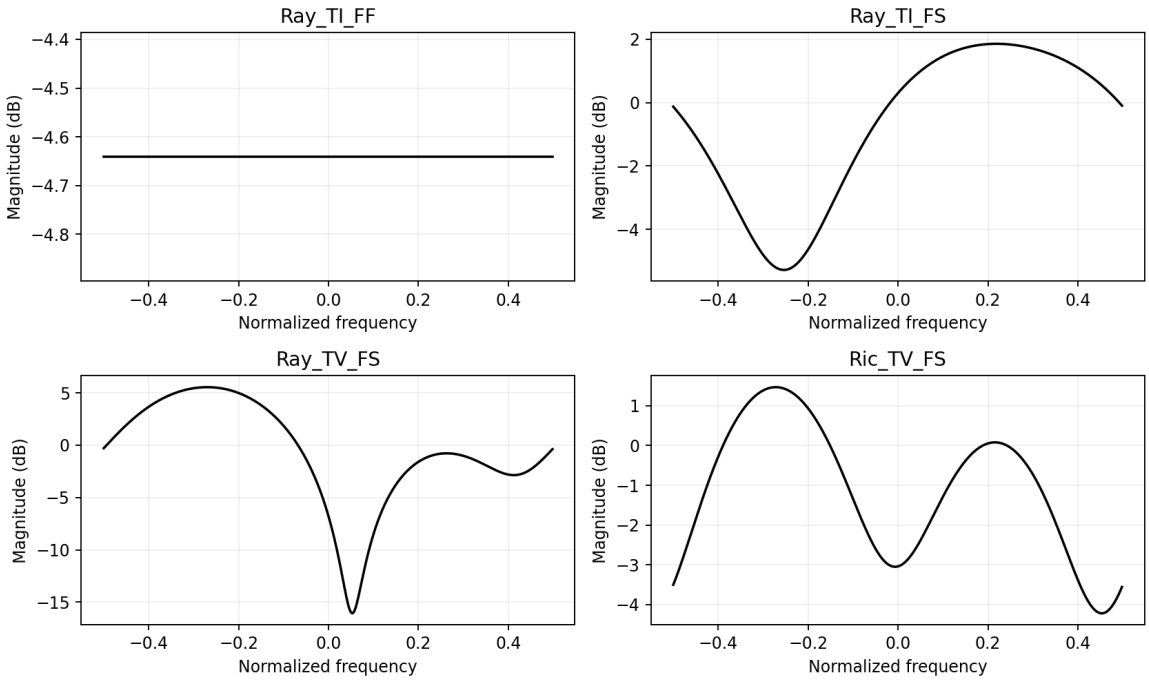


Figure 4: Frequency-response magnitudes at one observation time. A single flat-fading tap produces a frequency-independent response; delayed multipath taps produce frequency dependence.

With only one tap, the Fourier transform of the impulse response is constant in magnitude. With three distinct path delays, relative phases change across frequency and create a non-flat response. For time-varying channels, the plotted frequency response is a snapshot: there is no single time-invariant transfer function describing the entire observation interval.

4.5. Cumulative noise and received signal quality

Noise is added to all fading modes, not only to the AWGN reference. To isolate its contribution from propagation distortion, the reported noise-only error-vector magnitude is

$$\mathrm{EVM}_{\mathrm{noise}} = \sqrt{\frac{\sum_n |y[n] - s[n]|^2}{\sum_n |s[n]|^2}}. \tag{10}$$

This metric compares the noisy output with the same channel realization without noise. It is *not* a demodulated BER or a full equalized constellation EVM: it intentionally excludes distortion already present in $s[n]$.

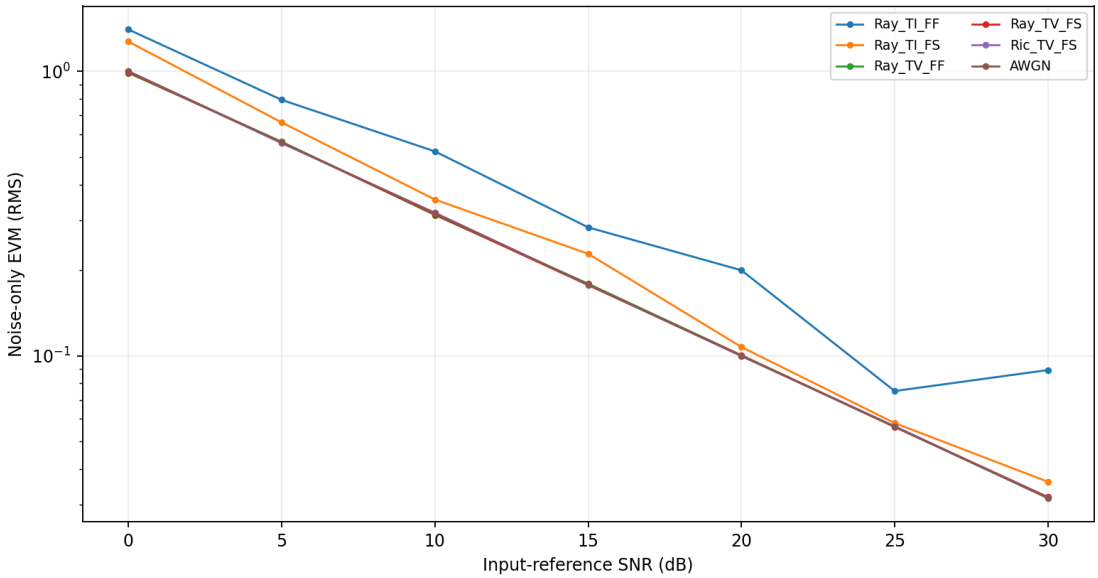


Figure 5: Mean noise-only EVM versus input-referenced SNR, from 20 independent realizations per channel and SNR point.

Across the six modes, higher input SNR reduces added noise. Channel-dependent differences remain because the denominator of Eq. (10) depends on the received signal power under each fading realization. This distinction prevents the misleading assumption that identical input SNR implies identical received quality under all propagation conditions.

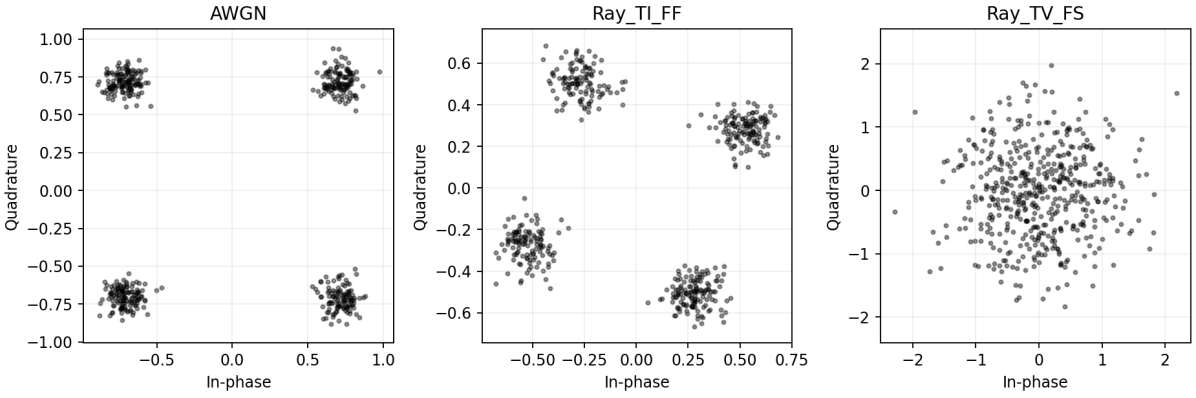


Figure 6: Uncompensated received QPSK samples at 20 dB input-referenced SNR. Fading and multipath distort symbol locations even where the additive-noise power is small.

In AWGN, QPSK clusters are centered near their transmitted constellation points. A static flat-fading channel rotates and scales the constellation for the duration of one realization. The time-varying frequency-selective case mixes symbols across delayed paths and therefore cannot be interpreted as an AWGN constellation without channel estimation and equalization. A BER comparison without specifying a receiver would not be meaningful, and no BER claim is made here.

5. Numerical Results

Table 4: Representative realization at 20 dB input SNR; measured powers are in normalized signal-power units.

Channel	Clean output power	Noise power	Measured SNR (dB)	Noise EVM
`Ray_TI_FF`	0.3435	0.0102	15.27	0.172
`Ray_TI_FS`	0.9744	0.0101	19.86	0.102
`Ray_TV_FF`	1.0610	0.0098	20.33	0.096
`Ray_TV_FS`	0.9590	0.0103	19.70	0.104
`Ric_TV_FS`	0.9920	0.0101	19.94	0.101
`AWGN`	1.0000	0.0098	20.10	0.099

The static flat-fading realization happens to experience attenuation: its received signal power is only 0.3435, causing a lower realized SNR despite the same input-referenced 20 dB setting. This is not a general ranking of the channel types; time-invariant Rayleigh fading can also produce stronger-than-average gains. The point of the table is to make input-side and receive-side SNR definitions explicit.

6. Implementation and Reproduction

The MATLAB function `simulate_channel.m` implements the six channel names in Table 1 and retains the required five-argument interface. The optional structure output makes the actual impulse-response coefficients, noiseless signal and additive noise accessible for channel analysis. The main script `run_channel_analysis.m` generates the six figure groups and three CSV result tables under `figures/` and `results/`. Run the script in MATLAB from any directory; its output paths are resolved relative to the script location.

The numerical experiments use a fixed, documented random seed for reproducibility *within the same implementation*. Since the model involves independent random channel and noise realizations, finite-record results vary when the seed or generator changes. The MATLAB code does not depend on a communications-specific toolbox, but the scripts should be executed in MATLAB for final environment-specific verification.

7. Conclusion

A common sampled complex-baseband model was developed for six requested channel conditions. Holding the channel taps fixed or allowing correlated temporal evolution distinguishes time-invariant and time-varying cases. A three-tap delay profile produces frequency-selective propagation, whereas the one-tap channel is frequency flat; a coherent LOS component distinguishes the Rician case. AWGN is modeled as additive noise, not as random channel taps. The numerical experiments show consistent time-correlation behavior and the expected reduction of noise-only EVM as input SNR increases.

These are controlled simulations with an illustrative power-delay profile, discrete path delays and an approximate Doppler model. They establish the behavior of the implemented channel simulator, not the measured performance of a particular wireless link. Receiver equalization and coded bit-error performance lie outside the specified scope.